

3.1.5 Common Built-in Functions (min, max, len, sum)

len () عناصر گنتا ہے، min () سب سے چھوٹا، max () سب سے بڑا، sum () اعداد کا مجموعہ، اور sorted () نئی ترتیب شدہ فہر...

CS XII

Chapter 3: Programming Fundamentals

Lecture 06 of 17

3.1.5 Common Built-in Functions (min, max, len, sum)

LEARN IT

Several built-in functions work across lists, tuples, sets and dictionaries. You do not call them with a dot on the object (except when a type also has its own method, such as `list.sort()`).

Function	Purpose	Works on
<code>min()</code>	Smallest element (numeric or alphabetical)	List, tuple, set, dictionary keys
<code>max()</code>	Largest element	List, tuple, set, dictionary keys
<code>len()</code>	Number of elements (for a dict: number of keys)	List, tuple, set, dictionary
<code>sum()</code>	Adds numeric elements	List, tuple, set, dictionary values via <code>sum(d.values())</code>
<code>sorted()</code>	Returns a new sorted list ; original unchanged	List, tuple, set, dictionary keys

```
s = {5, 1, 9, 3, 7}
print(len(s))
print(min(s))
print(max(s))
print(sum(s))
print(sorted(s))
```

OUTPUT

```
5
1
9
25
[1, 3, 5, 7, 9]
```

$5 + 1 + 9 + 3 + 7 = 25$. `sorted(s)` always returns a **list**, even when the input is a set.

WORKED EXAMPLE

The same five functions on four structures.

```
nums = [8, 3, 5, 3]
t = (8, 3, 5, 3)
s = {8, 3, 5}
d = {"math": 80, "eng": 70, "cs": 90}
print(len(nums), min(nums), max(nums), sum(nums), sorted(nums))
print(len(t), min(t), max(t), sum(t), sorted(t))
print(len(s), min(s), max(s), sum(s), sorted(s))
print(len(d), min(d), max(d), sum(d.values()), sorted(d))
```

OUTPUT

```
4 3 8 19 [3, 3, 5, 8]
4 3 8 19 [3, 3, 5, 8]
3 3 8 16 [3, 5, 8]
3 cs math 240 ['cs', 'eng', 'math']
```

On the dictionary, `min(d)` and `max(d)` compare **keys as strings**: "cs" comes first alphabetically, "math" last. `sum(d)` would try to add the keys "math" + "eng" + ... and fail. Use `sum(d.values())` — $80 + 70 + 90 = 240$. The set dropped the duplicate 3, so `len(s)` is 3 and `sum(s)` is 16.

sorted() versus .sort()

```
original = [8, 3, 5]
print("before sort:", original)
print("sorted():", sorted(original))
print("after sorted():", original)
original.sort()
print("after .sort():", original)
```

OUTPUT

```
before sort: [8, 3, 5]
sorted(): [3, 5, 8]
after sorted(): [8, 3, 5]
after .sort(): [3, 5, 8]
```

`.sort()` exists only on lists and rewrites the original. `sorted()` works on any iterable and leaves the original alone.

REMEMBER IT

LEN-MIN-MAX-SUM — count, smallest, largest, total. For a dictionary:

`len / min / max / sorted` use **keys**; `sum` needs `.values()`.

COMMON MISTAKE

`sum()` needs numbers. `sum(["a", "b"])` raises `TypeError`. Empty structures: `len([])` is 0, but `min([])` and `max([])` raise `ValueError` because there is no smallest of nothing.

اردو میں سمجھیں

`len()` عناصر گنتا ہے، `min()` سب سے چھوٹا، `max()` سب سے بڑا، `sum()` اعداد کا مجموعہ، اور `sorted()` نئی ترتیب شدہ فہرست دیتا ہے بغیر اصل کو بدلے ہیں۔ ڈکشنری پر `min` اور `max` کلیدوں پر چلتے ہیں؛ قدروں کا مجموعہ `sum(d.values())` سے نکالیں۔ `sort()` صرف فہرست کا میتھڈ ہے جو اصل فہرست بدل دیتا ہے۔

CHECK YOURSELF

1. For `s = {5, 1, 9, 3, 7}`, what are `len(s)`, `min(s)`, `max(s)`, `sum(s)`?
2. Does `sorted(s)` return a set or a list?
3. How do you total the values of `{"math": 80, "eng": 70, "cs": 90}`?

Answers: 1. 5, 1, 9, 25. 2. A list: `[1, 3, 5, 7, 9]`. 3. `sum(d.values())` → 240.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. `len(x)` works on:

- a. Sequences and many collections
- b. Only ints
- c. Only files
- d. Only NAND

MCQ 2. `min([3,1,2])` is:

- a. 1
- b. 3
- c. 2
- d. 6

MCQ 3. `sum((1,2,3))` is:

- a. 6
- b. 123
- c. Error always
- d. (1,2,3)

MCQ 4. `max` on a dict without a key function uses:

- a. Keys (careful!)
- b. Values always
- c. Lengths always
- d. Random

MCQ 5. These functions expect:

- a. Non-empty for min/max
- b. Always empty
- c. Only sets
- d. Only files

MCQ 6. `len("CS")` is:

- a. 2
- b. 1
- c. 0
- d. 3

Extra short questions

1. Four built-ins and a type they work on.
2. What happens `min([])`
3. sum of a list of strings?
4. len vs count in a list.

Extra long question

1. Given marks = [70, 80, 90] write expressions for n, total, mean, best, worst using built-ins. Show values. Warn about empty lists.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. len list; min tuple; max set; sum list of numbers.
2. ValueError.
3. TypeError — sum starts at 0.
4. len whole length; count(x) occurrences of x.

Long

1. n=len=3; sum=240; mean=80; max=90; min=70. Empty: don't divide; min/max error.
Link to Ch 4 statistics.

Next lecture: 3.1.6 Activity: Students Attendance Program